Preliminary build guide for V7.1:

Update A 21 May 2026

Update B 23 May 2026

This build guide is itself under construction; we're hoping to finalise it in early June

Please check the Web page for updates:

go to http://www.g1mra.com and select Resources then Designs

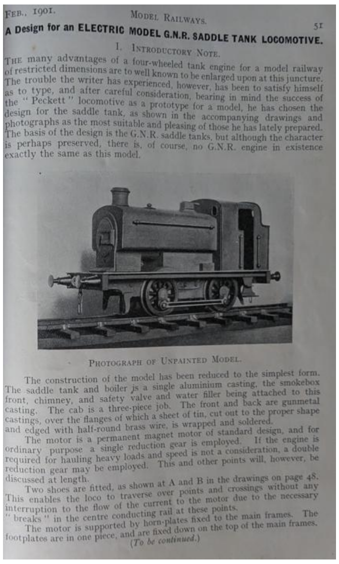
If in doubt, get in touch!

Adrian and David

David Viewing's Greenly 0-4-0 G1MRA starter engine

Background to the project

In the February 1910 issue of his magazine *Model Railways and Locomotives* Henry Greenly introduced a construction series for a simple 0-4-0 locomotive. He wrote '*The basis of the design is the G.N.R saddle tanks, but although the character is perhaps preserved, there is, of course, no GNR engine in existence exactly the same as this model'*. Henry is presumably referring to the GNR class J13-J17 0-6-0 saddle tank engines designed by Stirling.



FEB., 1901. MODEL RAILWAYS. 51

A Design for an ELECTRIC MODEL G.N.R. SADDLE TANK LOCOMOTIVE.

1. INTRODUCTORY NOTE.

THE many advantages of a four-wheeled tank engine for a model railway of restricted dimensions are to well known to be enlarged upon at this juncture. The trouble the writer has experienced, however, has been to satisfy himself as to type, and after careful consideration, bearing in mind the success of the "Peckett" locomotive as a prototype for a model, he has chosen the design for the saddle tank, as shown in the accompanying drawings and photographs as the most suitable and pleasing of those he has lately prepared. The basis of the design is the G.N.R. saddle tanks, but although the character is perhaps preserved, there is, of course, no G.N.R. engine in existence exactly the same as this model.

PHOTOGRAPH OF UNPAINTED MODEL.

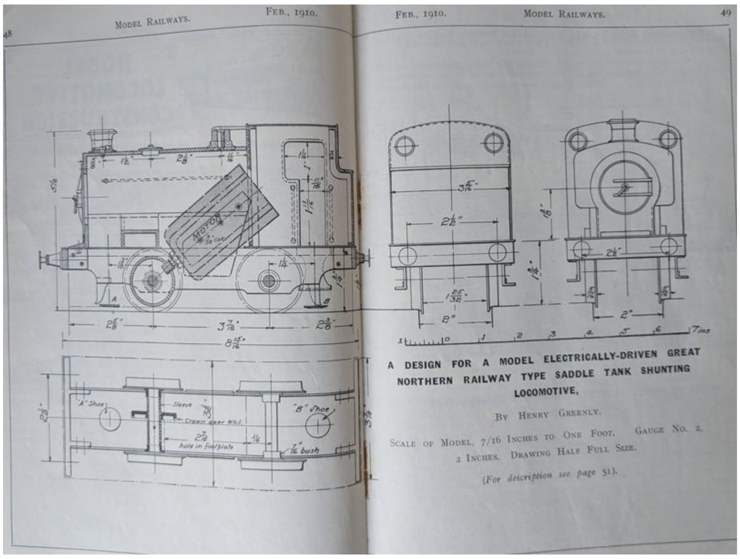
The construction of the model has been reduced to the simplest form. The saddle tank and boiler is a single aluminium casting, the smokebox front, chimney, and safety valve and water filler being attached to this casting. The cab is a three-piece job. The front and back are gunmetal castings, over the flanges of which a sheet of tin, cut out to the proper shape and edged with half-round brass wire, is wrapped and soldered.

The motor is a permanent magnet motor of standard design, and for ordinary purpose a single reduction gear is employed. If the engine is required for hauling heavy loads and speed is not a consideration, a double reduction gear may be employed. This and other points will, however, be discussed at length.

Two shoes are fitted, as shown at A and B in the drawings on page 48. This enables the loco to traverse over points and crossings without any interruption to the flow of the current to the motor due to the necessary "breaks" in the centre conducting rail at these points.

The motor is supported by horn-plates fixed to the main frames. The footplates are in one piece, and are fixed down on the top of the main frames.

(*To be continued.*)



Greenly's article finishes with the fatal words *To be continued* but as far as we know, there were no further instalments. The model as originally described is Gauge 2 (two inches between the rails) not Gauge 1. It seems that some models were sold commercially of which at least one survives, featuring an aluminium one-piece casting for the saddle tank. David Viewing owns the surviving example, and designed a radio control 3D printed chassis to go beneath it. He then developed bodywork in three sizes to go with it.

Here we see the whole family. From left to right: (i) the original Henry Greenly Gauge 2 body atop a modern chassis; (ii) in white, a Gauge 3 resin print; (iii) David's Gauge 2 prototype printed in black nylon with a coloured vinyl wrap; and (iv) on the right David's protoype Gauge 1 engine, printed in black nylon and sprayed with car-paint rattle cans.

The Gauge One Model Railway Association (G1MRA) has a long tradition of promoting 'starter' locomotives as an introduction to live steam, and recognised that a low-cost electric loco would also be an attractive way into large scale modelling. David found that Henry's simple design could be economically reproduced using commercial 3D printing technologies, and also researched low-cost radio control and even ultrasonic 'smoke' generators.

The prototypes were well-received by the G1MRA committee who then financially underwrote the production of ten pilot kits, with costs recovered from those builders who bought kits. At the 2025 AGM in Darlington, G1MRA awarded the 2025 President's Cup to David Viewing for this initiative, who then demonstrated the prototypes running at speed with a full load on the Midland Group's Carding Road track: you can watch a vide of that run at https://www.youtube.com/watch?v=jOOP2zo-2o0

Feedback from the ten pilot builders led to an improved V7.1 design in which the electronics were mounted in a separate backhead assembly to avoid water spillage from the smoke generator. At the same time, David added some detailing to the cab, and a flickering LED to simulate an open firebox.

This production version was formally launched with an article in the March 2026 edition of G1MRA's *Newsletter and Journal*. An initial run of twenty kits sold out immediately, with many collected at the Spring 2026 G1MRA meeting at Statfold Barn. Towards the end of the day, the Anglia Roads outer electric track was taken over by a procession of four trains, each hauled by a Greenly starter engine.

The lead train comprised eleven RCH wagons, a van and a brake van: this was easily handled by the engine.

Also on track was Elizbeth Scott's Cadbury loco hauling heavily weighted Northern Finescale vans, Steve Andrews' NCB loco and brake van, and Roger Hopkin's Archie.

The design, and the production process are well proven; the rest of this document comprises the build guide. Kits and built-up versions may be ordered from the G1MRA website at https://www.g1mra.com/ under the Sales/Models tab

Adrian Johnstone, April 2026

Build guide for the **V7.1** Gauge 1 Greenly starter engine

This update is ajGreenleyGuide7.1B dated 23rd May 2026

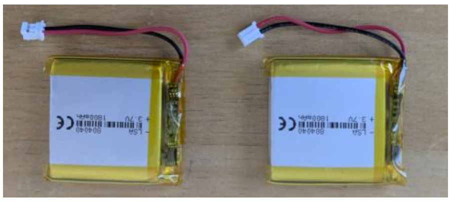
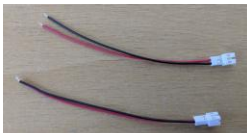
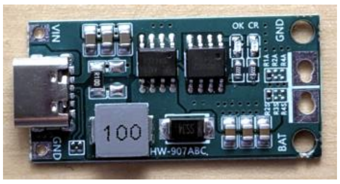
1: Check that you have a complete set of parts

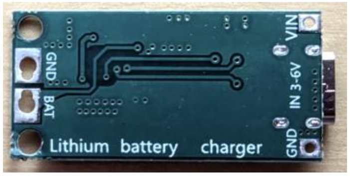
Here are the main elements of the kit as supplied.

Check that you have all of the following items

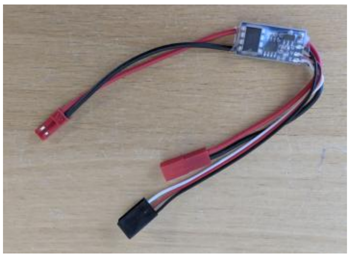
Note that some of the electronic components may already be wired together

Electronics

1. Two protected LIPO pouch cells.
   Treat LIPO batteries with great care! Do not puncture the pouches or short circuit the connections!
   
2. Two LIPO battery connectors and wires.
   Do not connect batteries to leads before they are soldered up as the loose leads may touch and short circuit the battery. Keep them separate for now.
   
3. Battery charger board fitted with a USB-C connector.
   

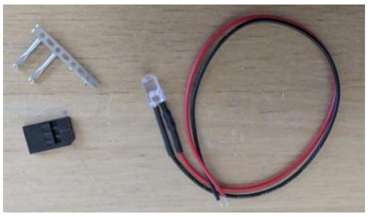
   

4. A single pole, double throw (SPDT) switch that connects the batteries to *either* the charger board or the electronic speed controller.
   

5. Electronic speed controller. The three-way Futaba connector plugs into the radio control receiver. The (socket) two-way connector at the same end as the three-way connector goes via the switch to the batteries; the other (plug) two-way connector goes to the motor.
   
6. Motor RS-385PH-2270 or RS385PV-2270.
    
7. Radio control receiver with attached short aerial lead, and a telemetry lead which will allow battery levels to be sent back to the hand-held transmitter. Keep these in the box for now.
   

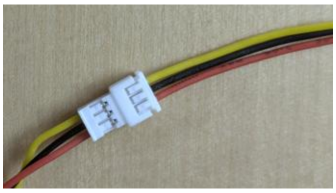
8. Hand-held radio control transmitter, boxed with manual, charging lead and wrist cord.


9. Flickering firebox LED with Futaba connector

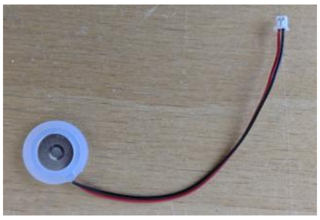


10. Hall effect sensor to control 'smoke' puffs

11. JST three-wire connectors for Hall effect sensor

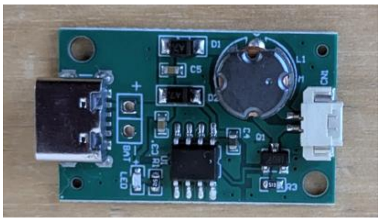
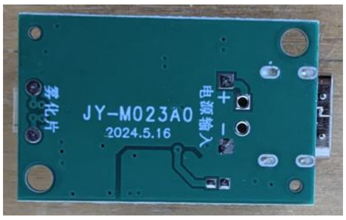


12. Ultrasonic 'smoke' nebuliser disc

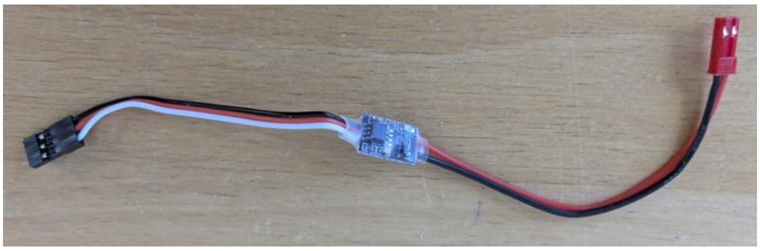


13. Nebuliser wick

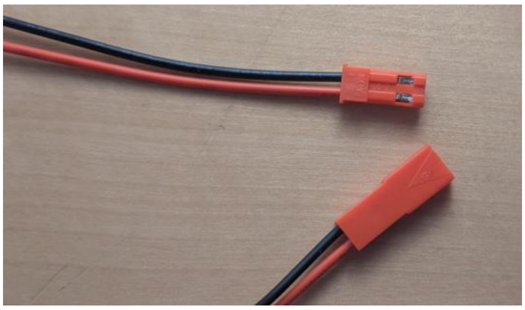
14. Ultrasonic control board – note this USB-C connector is not used

15. Ultrasonic interrupter switch

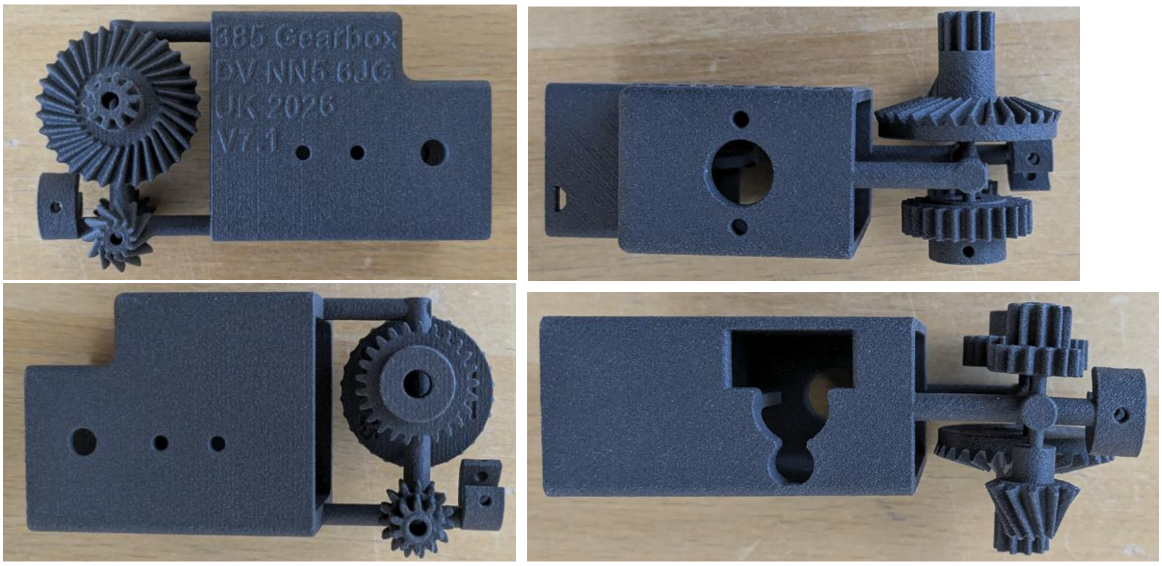


16. JST two-wire connector leads



Gearbox

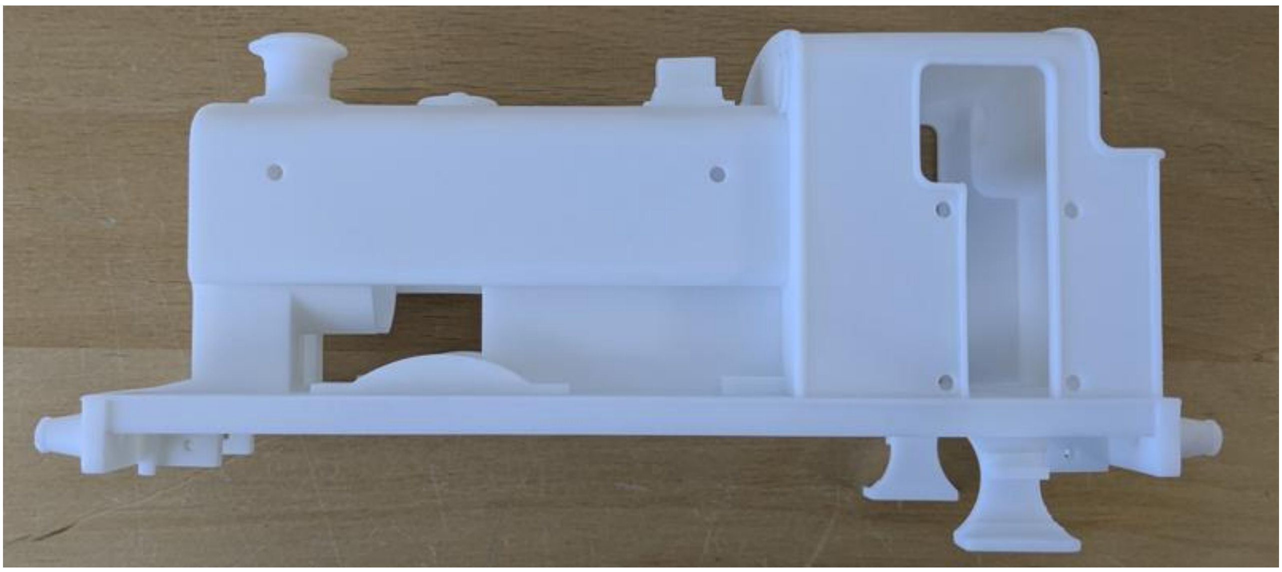
17. Printed nylon sprue with gearbox components



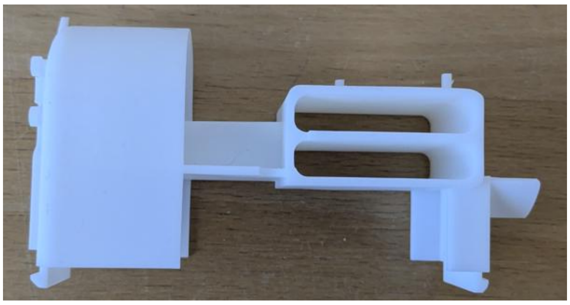
18. Two 5mm diameter axles, two 3mm diameter axles and one RZ-3 crown wheel bearing

Body and chassis

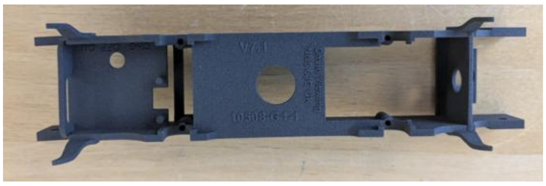
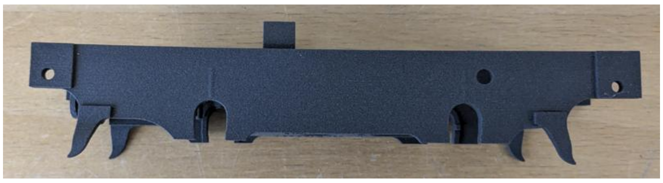
19. Printed body in white resin



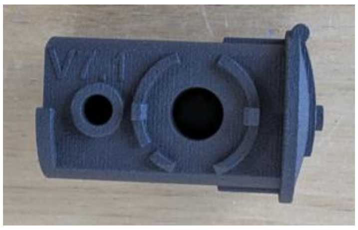
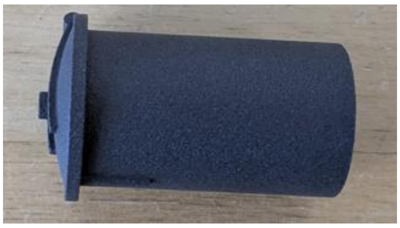
20. Printed backhead and electronics carrier in white resin

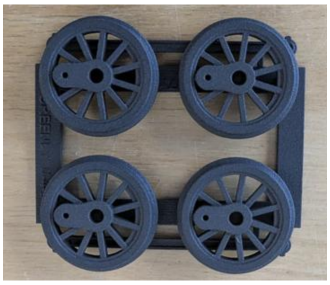
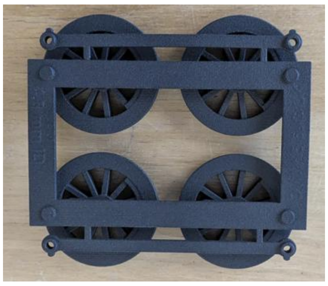
21. Printed black nylon chassis

22. Printed black nylon smoke generator and firebox door

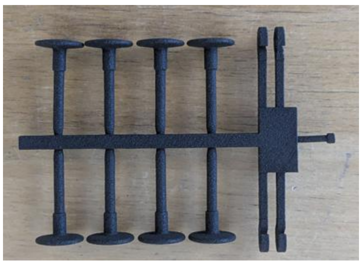
23. Printed nylon sprue with wheels and coupling rods

24. Printed nylon sprue with axle boxes and handrail knobs

25. Printed nylon sprue with further handrail knobs

26. Printed nylon sprue with buffers and coupling hooks



27. Four 2.5mm inside diameter buffer springs

28. One length (or four 10mm long cut pieces) of 2mm outside diameter spring for suspension

29. Two pieces of clock chain for three-link couplings

30. Four M2.5 8mm round head hex socket crankpins (coupling rod screws) with hex key

31. Four M2.5 buffer retaining nuts

32. Two M2.6 motor mounting screws

33. Four M3 slot head body fixing screws

34. Two 2mm diameter by 3mm Neodymium magnets

35. (Optional – not supplied) M2.5 by 6mm charging board fixing screw

(

36. Optional – not supplied) Two 2.5mm diameter by 4mm grub screws for final drive gear

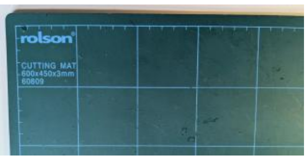
37. (Optional – not supplied) Steel or brass 1-2mm rod for hand rails

38. (Optional – not supplied) 180g of self-adhesive tyre balance weights for ballast

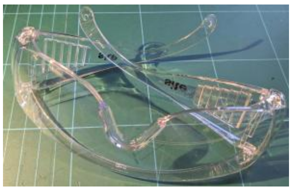
2: Collect tools

You will need some basic tools.

1. A cutting mat to protect your work surface

2. Safety glasses, particularly when cutting parts from sprues as they can fly off at speed

3. A 3mm slotted head screwdriver and a size 00 Philips driver

4. A robust craft knife, preferably with a retractable blade

5. A small electric drill or rotary tool

   I prefer the trigger operated style of electric drill as it gives me more control when I am opening out holes in plastic.

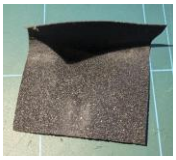
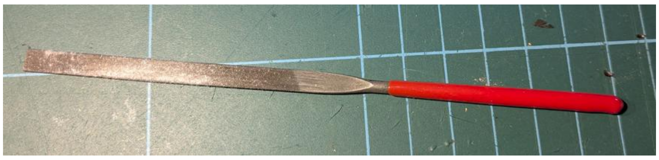
6. *Sharp* 2mm, 3mm and 5mm drill bits

7. Substantial side cutters for removing parts from sprues. I use a Xuron 2175 track cutter

8. Some emery paper or a small file to take the edge off of the pre-cut axles

9. A digital caliper may be useful to ensure that the right size screws are being used

If you are soldering the electronic connections, you will also need

10. Small side cutters

11. Small needle nose pliers

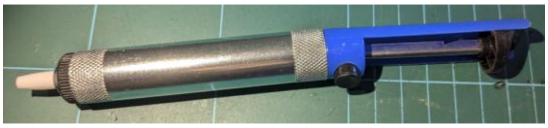
12. I find a pair of spring loaded tweezers useful for handling small components

13. A wire stripper is very helpful when removing insulation from thin wires

14. A soldering iron or soldering station, and some resin-cored solder

This yellow style of iron is common, but I recommend using a temperature controlled soldering station like the one on the right.

15. Mistakes happen, so a solder sucker for undoing joints is helpful



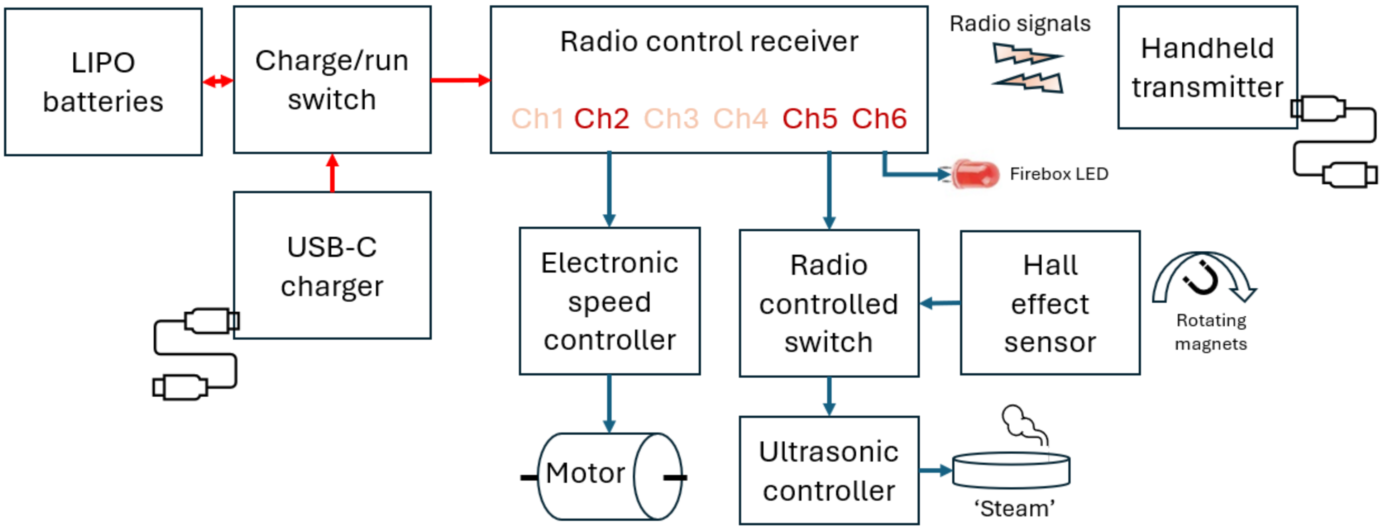
16. For connecting to open terminals on the switch, motor and LED, 2mm inside diameter heat shrink sleeving provides insulation and mechanical support.

17. After soldering, a hair dryer or heat gun is needed to shrink the sleeving.

3: Understand the electronic components

Outline of operation

This is a block diagram showing the electronic components and their interconnections.



Power

Power is supplied by two protected LIPO batteries wired in series to produce a nominal 7.5-8V supply. A charging board allows a USB-C power supply to be connected to the batteries, and a switch selects between charging and run modes. When not in use, the switch is left in the 'charge' position to avoid battery drain. A full charge should allow several hours of loco operation depending on load.

Radio control

The radio control system is based around the commercial HOT-RC DS600 radio control system which provides a cost-effective hand-held transmitter and receiver. The transmitter may be used with multiple receivers, but only one may be switched on at a time. Importantly, the controller provides 'cruise control' - pressing the button underneath causes the system to maintain that speed without needing joystick inputs: just what we need for continuous running. The transmitter has its own USB-C charging port; a full charge should give around 15 hours of transmitter operation.

The system provides six independent channels, three of which are in use in this loco. The joystick on the handheld controller sends a proportional signal to channel 1 for left/right motion and channel 2 for up/down motion. We only want to control motor speed, and use channel 2 so that up and down movement of the stick controls the engine's speed.

Channels 3-6 are on/off only, controlled by the four buttons arranged in a diamond on the controller. Channel 5 (the leftmost button) switches the smoke generator on and off. The other buttons are not presently used. Channel 6 on the receiver is used simply to supply power to the flickering firebox LED.

The system also has a single 'telemetry' back channel, which allows the transmitter to display the received signal strength and battery voltage at the receiver. If the receiver voltage approaches 7V, then it is time to put the loco on charge.

Motor control

The motor speed and direction are controlled by an Electronic Speed Controller (ESC), which is a standard piece of radio control equipment that plugs into the receiver and converts proportional inputs at the transmitter into pulses of electricity that drive the motor. The ESC has a three-wire connector called a Futaba connector that plugs directly into channel 2 of the receiver. It has a two-wire connection to the battery power, and a two-wire connection to the motor.

'Smoke' generator

The 'smoke' is really just a fine mist of water droplets, created by shaking a metal disk up and down at 108kHZ. A water tank under the disk and a wick are used to provide a supply of water to the mist generator. Left to itself, the ultrasonic controller board will excite the disk continuously, but we use a radio-controlled electronic switch connected to channel 5 to modulate the power supply to the ultrasonic board, and thus the output of 'steam'.

The electronic switch only powers the ultrasonics if (i) channel 5 is switched on *and* (ii) the Hall effect sensor detects a nearby magnet. The locomotive gearbox contains a wheel with two embedded magnets that rotate with the main drive axle, giving two pulses per revolution which then switch on the smoke generator for two quarters of each revolution, creating puffs of 'smoke'.

Firebox LED

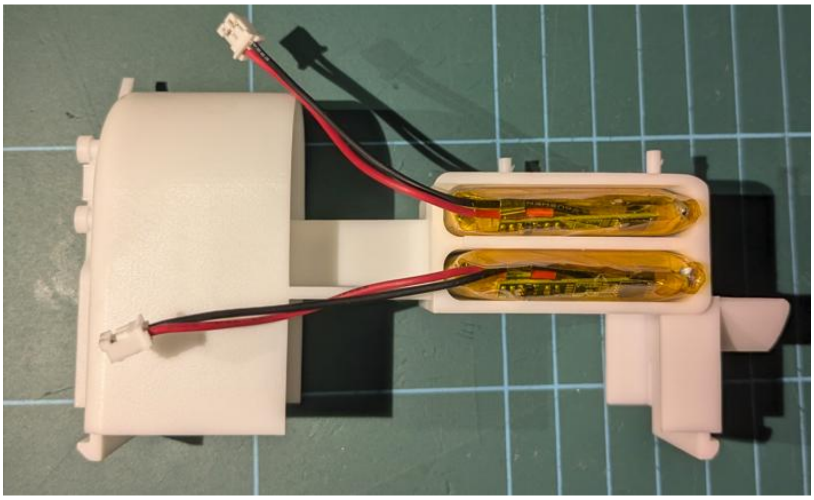
Red/orange light emitting diodes (LEDs) are available which have a built-in flicker. Simply by connecting one to the channel 6 power rail we create a firebox glow effect in the cab.

4: Initial physical battery check

LiPo batteries are remarkable devices but must be treated with respect. They must not be over-charged or over-discharged. Fortunately, the supplied batteries come with a small built-in circuit board which protects against the cell against over- and under-voltage conditions. They come already partially charged so if your batteries are not showing any charge at all then they are duds and should be replaced.

Do not at this stage connect the flying leads for the batteries as an accidental short circuit will damage the cells, and perhaps you too.

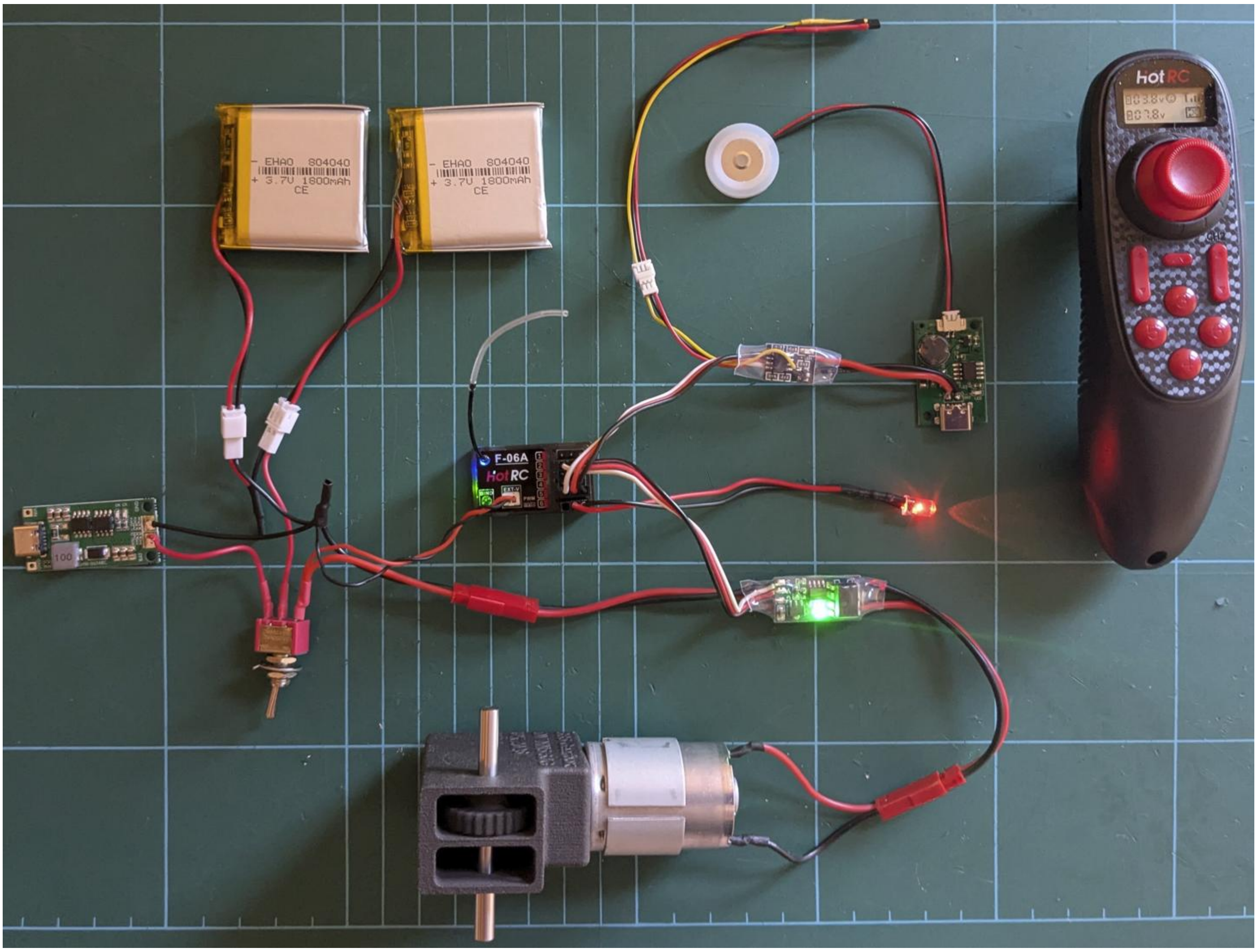
It is very important that the batteries are not physically damaged or swollen. Puncturing a cell can cause it to catch fire. The best way to check that the pouch cells are in good physical condition is to slot them into the backhead electronics carrier. The batteries should be an easy loose fit. If they are bloated and have to be forced in, then they should be replaced.

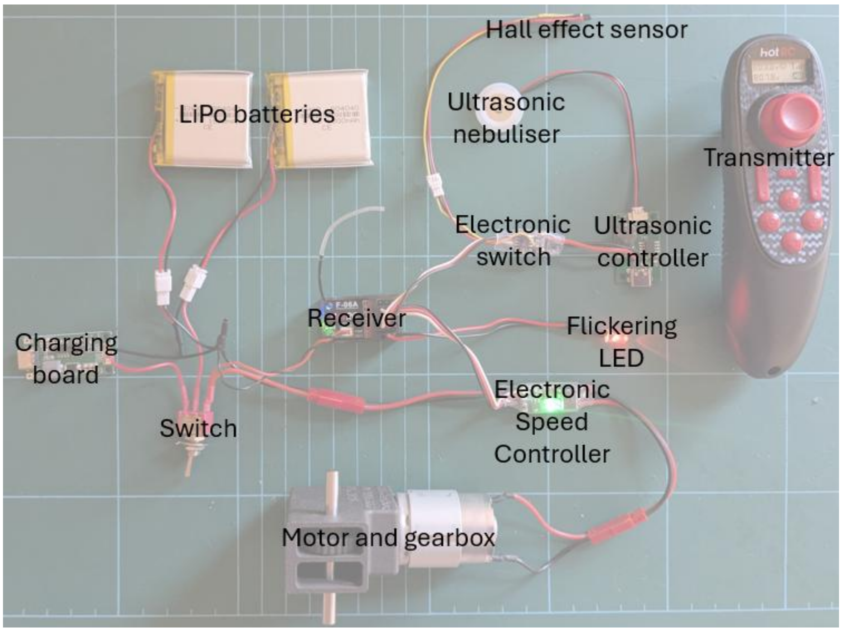


5: Bench test the electronics

The kit usually comes with pre-wired electronics ready to be plugged together as shown below.

If you are doing the soldering youself, then go through Appendix A before proceeding with this section.





Make sure that you carefully follow the picture to ensure everything is properly connected.

You may find it useful to zoom into the picture to get a closer look at the connections.

Note that there are two sets of 2-pin JST connectors wired to the Electronic Speed Controller (ESC) The socket goes to the switch for battery power; the plug goes to the motor.

The ESC connects to channel 2, the ultrasonics to channel 5 and the LED to channel 6

Please also view this video to see the electronics in action: https://www.youtube.com/watch?v=vV_OIAuo9Q0

You can test the electronics with the motor loose on the bench, or after assembling the motor into the gearbox as described in section 7.

Once you have everything connected, make sure that the transmitter is switched off, and then switch on the receiver with the toggle switch. Just as in the picture above, you should see a continuous blue light on the receiver and a green light flashing on the receiver at approximately 1Hz. You should also see the firebox LED glow orange with occasional changes in brightness, and the green light on the Electronic Speed Controller should flash with a double beat.

Now switch on the transmitter using the small central button immediately below the joystick. The transmitter will play a short ascending tune, and the flashing green lights on both the receiver and the ESC should stop flashing and stay on continuously.

If this change in the green lights does not happen then you need to 'bind' the receiver to the transmitter so that they recognise each other.

Binding the receiver to the transmitter

Binding is performed by starting both the receiver and the transmitter in a special mode, by holding down buttons as they are switched on. The receiver has a button labelled Bind just below the antenna connection, and the transmitter has a large red button underneath labelled OK. We call this the cruise control button.

Switch off both the receiver and the transmitter. Now hold the Bind button on the receiver down and switch on with the toggle. The green light will flash rapidly to indicate that it is listening for a new transmitter. Now, whilst pressing the cruise control button underneath the transmitter, switch the transmitter on.

The transmitter will play its rising tune, and then beep repeatedly. Press the cruise control button again and the beeping will stop, the receiver blue light will go out, and the receiver green light will stop flashing. You have now bound the receiver and the transmitter.

You only need to do this once: the receiver will remember which transmitter it 'belongs' to. If you have more than one engine, you can bind them both to the same transmitter, but you will only be able to use one at a time.

Testing the motor and setting the zero point

Switch on the transmitter and receiver. You should now be able to run the motor forwards and backwards using the joystick. If the motor hums or turns when the stick is at its central 'off' position, use the trim buttons on the transmitter to set a zero point.

Using the cruise control

Set the motor running by pushing the joystick and then press the cruise control button and release the joystick. The motor should continue to run at the same speed, which is very helpful when running on a continuous loop.

Charging

Once you have the motor turning under the control of the transmitter you can test the charging board by connecting a USB-C charger. The charger board will show a red LED when charging the battery which turns blue when the battery is fully charged. A full charge takes 4-5 hours, which will then run the engine for several hours depending on load.

Telemetry

Check the voltage display on the transmitter to ensure that the telemetry connection is working. When fully charged it will show around 8V, reducing to about 7V when the battery is nearly flat.

Ultrasonics

We delay testing the ultrasonic smoke generator until the rest of the engine has been assembled.

6: De-sprue printed parts and clean up

Prepare axles

The axles need a bevel on the ends so that sharp corners do not tear the plastic parts as they are pressed on. The bevel also helps provide some 'lead' when inserting. This can be done by spinning the axle in a drill with the end against a piece of emery paper, or by careful work with a file.

Cut springs

If not supplied already formed, cut four 10mm lengths of the 2mm outside diameter spring for the axle boxes, and four 15mm lengths of the 2.5mm inside diameter spring for the buffers.

Cut and clean up gear box components

Before removing from the sprue, we need to ream the holes in the crown gear and the idler fear so that they may rotate freely on their fixed axles. The nylon material has a tendency to smear and tear, so rather than using a proper reamer we recommend drilling out the holes with a *sharp* 3mm bit. It is best to do this whilst the gears are still on the sprue to make them easier to handle, but it can be done with care after cutting.

Now remove all items from the sprue.

Trim the stubs of sprue from the gears since these might touch the frame and create noise, or bind against the frame.

The idler gear as printed is a little oversized. Trim the smaller gear end with a craft knife until the gear slides freely between the narrow walls of the gearbox.

Cut and clean up axle boxes

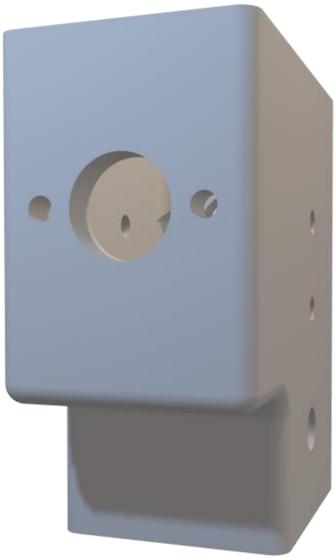
Before removing from the sprue, clean up the holes using a sharp 5mm drill. Then cut from the sprue and clean up the stubs.

Clear charging socket cutout

7: Gearbox assembly

Here is an exploded diagram of the gearbox.

Under construction in Powerpoint using STLs



The elements of the gearbox should be assembled in this sequence:

1. Motor
2. Motor bevel pinion
3. Idler gear
4. Crown wheel
5. Final drive gear
6. Timing wheel with magnets
7. Driver axle boxes
8. Driver wheels

The idler gear and the crown wheel rotate freely on their fixed axles. The motor bevel pinion, final drive gear, timing wheel and driver wheels are rigidly fixed to their axles which rotate

Motor

Begin by attaching the motor to the gearbox frame using the M2.6 screws.

Motor bevel pinion

The motor pinion is an interference fit, and some force may be needed to get the pinion onto shaft.

Warning: There is a danger that the motor internals might be damaged if you force the pinion on whilst holding the motor body. Instead, stand the motor up on a solid surface so that the end of the shaft is supported, and push the pinion on with force being transmitted through the shaft to the surface and not though the body of the motor. It is a good idea to reconnect the motor to the electronics after fitting the pinion to make sure everything is still working correctly before proceeding.

Idler gear

Ensure that you have trimmed the smaller gear end with a craft knife until the gear slides freely between the narrow walls of the gearbox.

The idler should spin freely (but without wobbling) on a 3mm axle. Open the bore out with a drill as necessary.

Slide the idler gear into the box, making sure that the smaller gear is against the external wall of the gearbox and the bore is aligned with the sidewall holes, and then push the axle through. Support the inner frame with a small block of wood to prevent bowing as the shaft is inserted.

Crown wheel

Snap the ball race bearing into the base of the crown wheel and align the wheel with the axle holes in the gearbox housing. The crown wheel should engage with the motor pinion and the idler gear.

Insert the axle from the side of the gearbox away from the bearing. I found it helpful to put the box on the cutting mat on its side (crown wheel down) and gently tap the axle down the last few mm to seat it in the wall of the gearbox. You might prefer to use a vice.

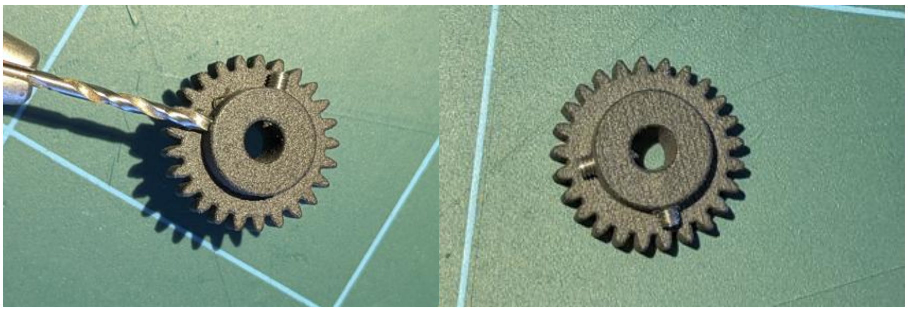
Check the alignment of the motor pinion by manually turning the crown wheel and then running the motor under power. Ideally the pinion should mesh tightly with the crown wheel but not so tightly as to cause binding: there should be a little backlash. Adjust the alignment by slight movements of the motor pinion up and down the motor shaft – a pair of long nosed pliers or a flat bade screwdriver will help.

Final drive gear

The final drive gear is an interference fit onto the 5mm main axle. As before, I put the gearbox crown-wheel down onto the cutting mat and drove the axle through the final drive gear and gearbox wheel with gentle taps from a small hammer. You might prefer to use a vice.

Warning: The drive gear must be a tight fit on the axle since this carries drive between the motor and the wheels.

There are holes for M2.5 grub screws if your gear is loose; drill out the holes to 2mm and carefully drive the screw in – it will self-tap.

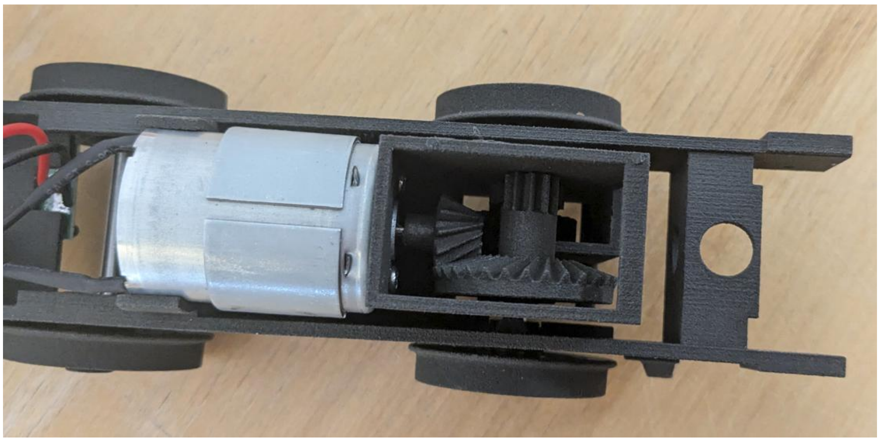


Alternatively, put a drop of super glue down the hole and rotate the axle through the gear to spread the superglue round the inside of the glue. **Only do this after the wheels have been fitted** (see Step 7) since once glued you will not be able to adjust the position of the gear on the axle.

Running in

Run the gearbox on the bench for a while to get everything bedded in, and then check alignments. Do not be tempted to add any sort of lubricants as the nylon gears do not need them – all you will do is create a sticky dust-collecting mess.

Warning: The crown wheel axle and the idler axle (the 3mm ones) should be gripped by the walls of the gearbox and should not turn. If the axles do rotate, then they will in time work their way out and the gears will come out of alignment, and probably be damaged. After running in, if you see that they are moving, then let a couple of drops of superglue seep into the axle/wall interface and leave to set



8: Chassis assembly

The chassis clips together in an ingenious way: the motor goes into the round bracket on top, and the sprung axle boxes clip into the frames. Everything can be unclipped, so you can safely try a dry run before final assembly.

Adding wheels and axle boxes to the main axle

Add an axle box to one side of the main axle. The flange on the axle box should on the outside – it provides a rubbing surface for the wheel.

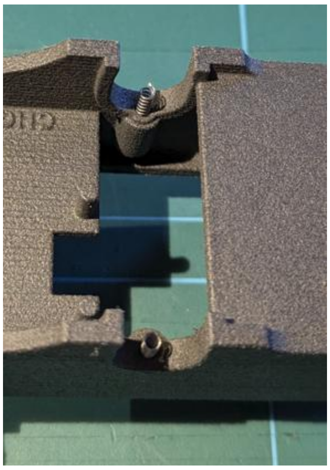
Now press on a wheel. This requires considerable force, and the wheel needs to be set accurately at a right angle to the axle – it is easy to press the wheel on at a slight angle so you end up with a wobbly wheel. (Guess how I know.) The best way to do this is with a vice: do not attempt to do it by hand. Check the alignments carefully with a set-square, and drive the axle in slowly by tightening the vice jaws, checking as you go.

Now repeat for the other side of the main axle, and then make up the undriven axle in the same way.

Fitting the motor and gearbox assembly

Easing the motor through the frames requires some care as there are several lumps and bumps on the inside of the frame that can block the motor's passage. Insert the top of the motor from below, and slide it around to find the widest part of the frames, then ease the frames apart slightly to allow the motor waistband through.

Fitting springs and axle boxes



Invert the frame and drop springs into the spring pockets for the axles. Rest the driven axle and its axle boxes on the top edge of the frame and then rotate both axle boxes until the indent for the top of the spring is facing down. Now push the assembly down – it should mate with the frame with a satisfying click, with the springs providing suspension. If all did not go well, unclip the axle boxes from the frame and try again.

Repeat this with the undriven axle, and seat the motor properly in its clip.

Place the chassis right side up, and check that the springing is effective.

9: Bodywork, buffers and ballast

The white resin body will take rattle-can paint directly, but a first coat of car body primer coat is advisable. Some livery suggestions are shown in the next section.

The engine is mostly plastic, and thus very light. There is plenty of power to pull heavy loads, but the engine needs mass to achieve the necessary adhesion and avoid wheel spin. Adding about 180g of ballast is recommended, most easily achieved by adding self-adhesive tyre balancing weights. Alternatively, a piece of lead flashing may be folded into the tank interior.

Handrail knobs maybe superglued into the holes in the body (which may need opening out). They are designed to take 2mm wires which may look a little chunky. Some builders have used 1mm or 1.2mm rodding which is closer to a scale size with a little dab of glue to keep them in place.

Nylon buffers and coupling hooks are supplied. The buffers can be glued permanently in place, or you can add the supplied springs with the buffer nuts tightened directly onto the buffer shanks. You will need to open out the buffer shanks with a 4.1mm drill to allow free movement.

Serving suggestions – preproduction run

David Viewing's rattle-can painted and vinyl wrapped prototypes in action at the 2025 G1MRA AGM

Steve Andrews' weathered NCB loco with added external cylinders, safety valves, bunker and spectacle bars

William Ayerst's completed engine and Roger Hopkin's Archie

Elizabeth Scott's Cadbury engine running on Mike and Christine Blands' track with Northern Finescale vans

Serving suggestions – larger than Gauge 1

David Viewing's Gauge 3 prototype running on the Warton Road layout at the 2026 National Garden Rail Show

The surviving GNR J13 preserved in the National Collection

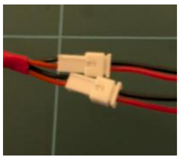
Appendix A: Soldering the electronics

We assume that you have the necessary skills to solder electronic components; if not then please purchase the ready-soldered version of the kit.

Rest of this section is under construction

General advice

We recommend using heat shrink tubing for connections to the switch, the motor and the LED so as to provide some mechanical support for the wires and insulation against the elements – Gauge 1 tracks are rather hostile environments. You will also need wide heat shrink sleeving to re-encapsulate the RC switch after soldering.

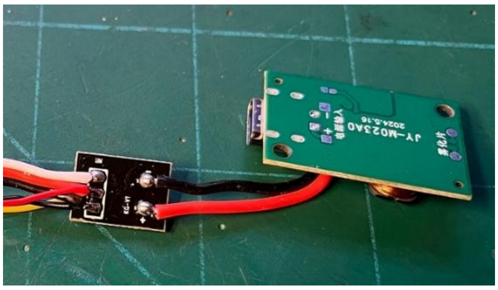


Warning: connectors are keyed, but can have the red and black leads connected either way round, which is unhelpful. To avoid nasty surprises, always connect red to red and black to black; if the connector leads do not match up, then ease the pins out of the shell and rearrange like this.

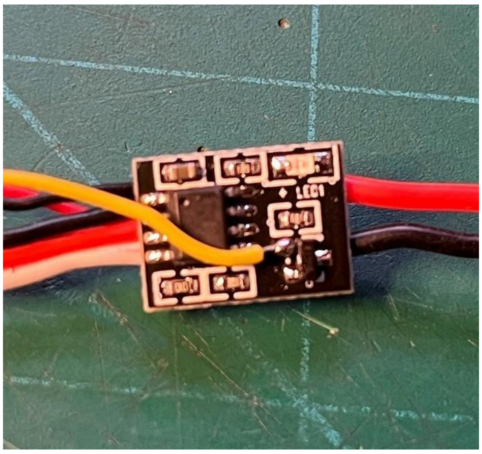
Modifying the electronic switch.

The sensor generates 'puffs' by shutting off the output of the RC switch as the magnet wheel rotates. It does this by shutting off the output transistor in a 'cluge' which has been shown to work reliably.

First , carefully peels back the Heatshrink covering from the switch pcb taking care not to damage any of the components. Prepare the JST-PH 1.25mm connector so the yellow wire is approx 30cm long, naking the red and black wires slightly shorter. Bare the insulation for 1-2mm and pre-tin the ends. Attach the black and red wires by soldering the on top of the existing wires on the board.



Now, pre-tin the transistor pin as indicated and attach the yellow wire taking care not to short to anything else.



Re insulate the board with transparent heatshrink 22mm wide (not supplied).

Magnetic sensor (A3144) detects the passage of magnets rotating past on a special 'magnet wheel' installed in the bottom of the gearbox. Positions for 4 magnets at 90 degreess are provided for the 2 -cylinder engine but in practice at model railway speeds the the smoke generator quickly gets 'swamped' by the rate of 'chuffs'. A better compromise is to install just two magnets in opposing positions, which gives a detectable 'chuff' up to a reasonable forward speed.

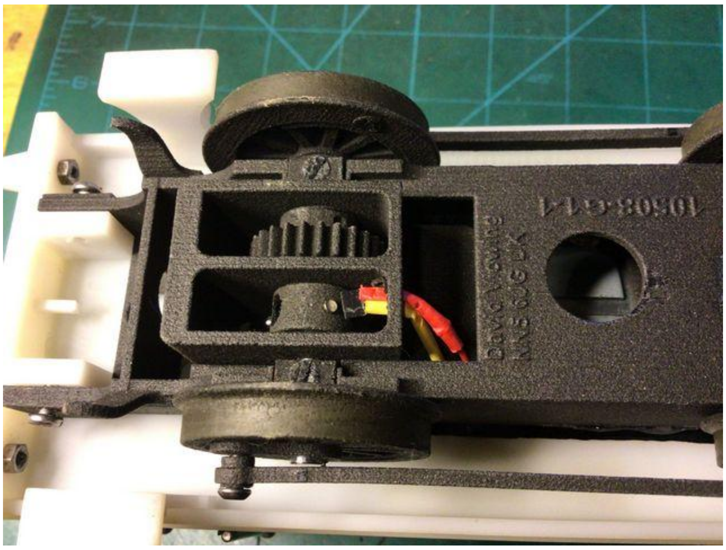
Magnet Polarity

The magnet polarity is important because the sensor works in one way only. The chamfered side of the sensor should face the magnet's South Pole. You can find this by trial and error, unless you happen to have a compass! Both magnets must have the same polarity facing outward.

Once you have established the correct polarity, push the magnets into two opposing holes in the magnet wheel until about 1mm projects. I find the best way to do this is to pick up the magnet with a wide flat screwdriver blade or similar. The magnet wheel clips onto the driven axle and can be removed at any tine for alteration if required.

Sensor Installation

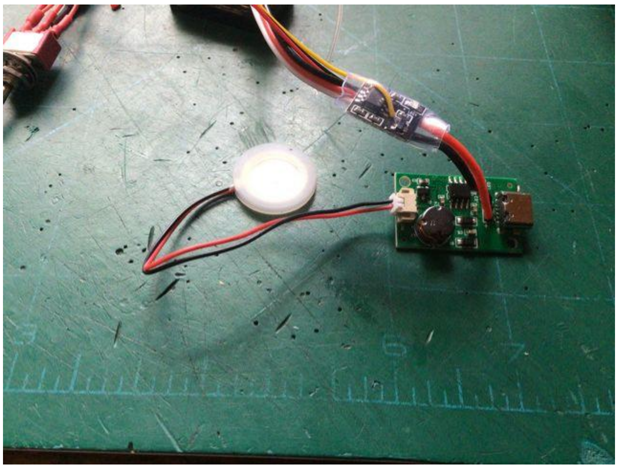
Once the three wires from the female JST connector (The one with receptacles for the pins) have been attached and insulated, The sensor can be inserted in the bottom flange of the gearbox shell. The trapezoidal cutout will need little easing with a sharp knife and provided this is not overdone the insulation will hold the sensor in place. Insert the sensor chamfered face down, facing the rotating magnets and as close as practicable without them hitting.

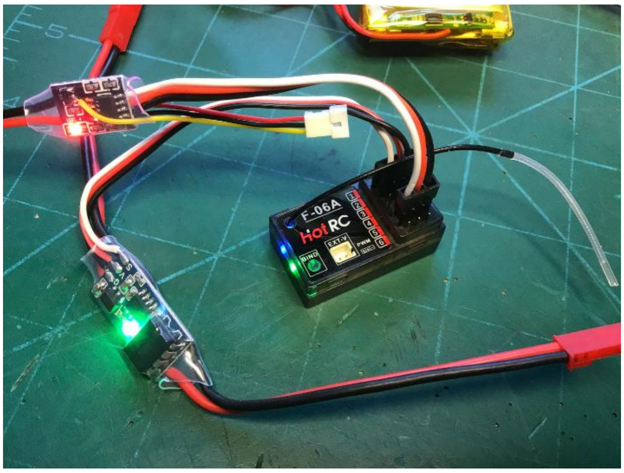


Connecting the Ultrasonic oscillator

To save space, we connect the RC switch directly to the smoke unit oscillator. This means cutting off the JST connector leaving about 20cm of wire projecting. The connector can be re-used as part of the on/off switch assembly.

Pre-tin the ends of the remaining wire stubs after removing about 2mm of insulation. Also pre-tin the two pads on the oscillator board, immediately behind the connector, noting they are marked + and -.. Solder the RC switch output leads to these pads, observing red goes to +, etc.

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